



Django

A Web Framework with Immense Potential

In the era of cloud computing, Web frameworks have great importance in a Web developer's life. Choosing the right one is crucial to the success of a Web application. This article explores one such (not so) popular Web framework—Django.

The success of a Web application depends on factors like managing high traffic, maintaining site code, quick updates, bug tracking, ready-made tools, code re-usability, help for code development, etc – all of which become important in a Web framework. There are a lot of Web frameworks available that have been written in various programming languages. In reality, only a few are used extensively and are recommended by the industry. So what is so special about those few?

Django is a framework developed for *Lawrence.com* by Simon Willison and Adrian Holovaty. It is written in Python. Every Web framework is an effort to counter identified loopholes in an existing technology. Django was developed for the same reason. Initially, at *Lawrence.com*, code was being written in PHP, but code maintenance was a big headache for developers. That's when Willison and Holovaty, both fond of Python, decided to write a Web framework based on Python. One of the basic features of Django is that it's a URL-driven Web framework. Developers could design their own URLs, and choose what was to be shown for that URL, quite unlike the frameworks existing at the time.

As I mentioned earlier, when it comes to choosing the right Web framework, there is now a 'problem of plenty'. Since there are many options, comparison is inevitable. I do not

intend to make this article a battle between Django and other frameworks. I would rather aim at discussing a few points that forced me to use Django as our core Web framework for AppSurfer (<http://www.appsurfer.com>).

1. **Coding standards:** Code readability is one of the most important aspects of software development. It becomes crucial when more than one person works on a code piece. Python believes in 'The One True Way'—i.e., there should be a single way of writing any code. Therefore, rather than two or three different options on syntax, method calls, etc, there's usually one option in Python.
2. **Python:** Django is based on Python, and hence benefits from all the advantages of Python. Python has a clean and elegant syntax, and is supported by a large library of standard and contributed modules, which cover everything from multi-threading to the zipping of files. The language's object-oriented model is especially suited for MVC-style development. Sooner or later, performance will become a major concern with Web projects, and Python's runtime environment shines here, as it is known to be fast and stable. Python supports a wide range of Web servers through modules, including the infamous Apache. Furthermore, it is available for all the major platforms: UNIX, Linux, Windows, and Mac. Python also supports a